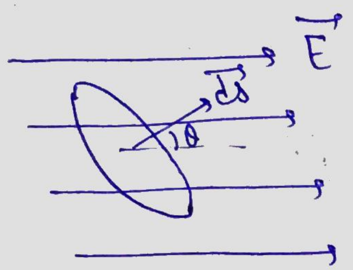
Electric flux (ϕ)

The No. of Electric field lines passing through the given area.



$\phi = E\,ds\cos\theta$

$$\boxed{\phi = \vec{E}\cdot\vec{ds}}$$

θ is angle b/w $\vec{E}$ & $\vec{ds}$

Scalar Quantity

* $\boxed{\text{S.I unit } Nm^2/C}$

Continuous charge distribution

① Linear charge density (λ)

$$\boxed{\lambda = \frac{dQ}{dl}}$$ charge per unit length.

② Surface charge density (σ)

$$\boxed{\sigma = \frac{dQ}{ds}}$$ charge per unit Area.

③ Volume charge density (ρ)

$$\boxed{\rho = \frac{dQ}{dV}}$$ charge per unit volume.